

Microprocessors and Microcontrollers

Assignment 12- Week 12

TYPE OF QUESTION: MCQ

Number of questions: 15

Total mark: 15 X 1 = 15 M

QUESTION 1:

Correct Answer: c

Detailed Solution:

Step 1: Convert Binary to Decimal

The given binary number $(10100000)_2$ corresponds to:

$$\begin{aligned} & (1 \times 2^7) + (0 \times 2^6) + (1 \times 2^5) + (0 \times 2^4) + (0 \times 2^3) + (0 \times 2^2) + (0 \times 2^1) + (0 \times 2^0) \\ &= (1 \times 128) + (0 \times 64) + (1 \times 32) + (0 \times 16) + (0 \times 8) + (0 \times 4) + (0 \times 2) + (0 \times 1) \\ &= 128 + 32 = 160 \end{aligned}$$

Step 2: Compute the Output Current

The DAC0808 uses an 8-bit resolution, so the output current is given by:

$$I_{out} = I_{ref} \times \frac{D}{256}$$

where:

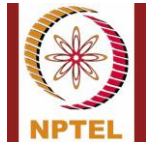
- D is the decimal equivalent of the binary input.
- 256 is the maximum possible value for an 8-bit DAC (since $2^8 = 256$).

Substituting the values:

$$I_{out} = 5mA \times \frac{160}{256}$$

$$I_{out} = 5mA \times 0.625$$

$$I_{out} = 3.125mA$$



QUESTION 2:

Correct Answer: a

Detailed Solution:

The 8086 microprocessor can operate in two different modes:

1. Minimum Mode: Used for small, single-processor systems.
2. Maximum Mode: Used for systems with multiple processors or coprocessors.

The MN/MX (Minimum/Maximum) pin is responsible for selecting between these two modes:

- $MN/\overline{MX} = 1 \rightarrow$ Minimum mode (Single processor system).
- $MN/\overline{MX} = 0 \rightarrow$ Maximum mode (Multiprocessor system).

Thus, the MN/MX pin determines the operating mode of the 8086 microprocessor.

QUESTION 3:

Correct Answer: c

Detailed Solution:

The Intel 8086 microprocessor has a 20-bit address bus, which allows it to address up to:

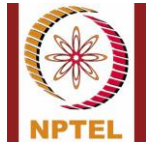
$2^{20} = 1,048,576$ (1MB of memory)

Even though the 8086 is a 16-bit processor, its segmented memory addressing scheme enables it to use a 20-bit address bus to access 1MB of memory space.

QUESTION 4:

Correct Answer: d

Detailed Solution:



❑ (i) 16-bit data bus → 8086

- The 8086 processor has a 16-bit data bus, meaning it can transfer 16 bits of data at a time.

❑ (ii) 8-bit data bus → 8088

- The 8088 processor has an 8-bit external data bus, even though internally it is a 16-bit processor. This design was chosen to make it compatible with 8-bit peripheral hardware.

❑ (iii) BHE (Bus High Enable) signal → 8086

- The BHE (Bus High Enable) signal is used in the 8086 processor to indicate whether the high byte of the 16-bit data bus is being used.
- Since 8088 has an 8-bit data bus, it does not need BHE.

❑ (iv) 20-bit address bus → 8086

- Both 8086 and 8088 have a 20-bit address bus, allowing them to access 1MB of memory.
- However, since most options in the question pair (iv) with 8086, we select 8086 here.

QUESTION 5:

Correct Answer: c

Detailed Solution:

In the 8086 microprocessor, the CX (Count Register) is specifically used as a loop counter in loop and iteration instructions.

- The LOOP instruction in assembly language automatically decrements the CX register and checks if it reaches zero. If not, the loop continues.
- It is also used in string operations and shift/rotate instructions as a counter.

QUESTION 6:



Correct Answer: d

Detailed Solution:

The 8086 microprocessor is divided into two main units:

1. Bus Interface Unit (BIU) – Responsible for fetching, reading, and writing data from/to memory and I/O.
2. Execution Unit (EU) – Responsible for decoding and executing instructions.

Components of the BIU:

- Instruction queue → Present (used for instruction prefetching).
- Segment registers → Present (CS, DS, SS, ES for memory segmentation).
- Instruction pointer (IP) → Present (holds the address of the next instruction to be executed).

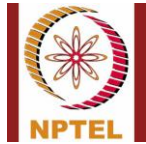
Component NOT in the BIU:

- Flag register → NOT present in BIU
 - The flag register is part of the Execution Unit (EU) and is used for status and control flags.

QUESTION 7:

Correct Answer: b

Detailed Solution:



The physical address in the 8086 microprocessor is calculated using the formula:

$$\text{Physical Address} = (\text{Segment} \times 16) + \text{Offset}$$

Given:

- Segment = 7213h
- Offset = 5686h

Step 1: Convert Segment to a Physical Address Base

Multiply the segment value by 16 (or shift left by 4 bits):

$$7213h \times 10h = 72130h$$

Step 2: Add the Offset

$$\text{Physical Address} = 72130h + 5686h$$

Now, perform the addition:

$$72130h + 5686h = 777B6h$$

QUESTION 8:

Correct Answer: b

Detailed Solution:

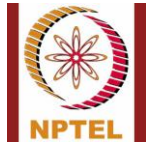
In the 8086 microprocessor, some address lines are multiplexed with other signals to reduce pin count and improve efficiency.

- Address lines A19 – A16 are multiplexed with status signals (S6 – S3).
- These status signals provide additional control information about the current operation of the processor.

Breakdown of Address Line Multiplexing in 8086:

1. A0 – A15 → Used as a dedicated address bus. (Not multiplexed)
2. A16 – A19 → Multiplexed with status signals (S3 – S6).
3. AD0 – AD15 → Multiplexed with data bus (D0 – D15).

QUESTION 9:



Correct Answer: b

Detailed Solution:

The 8086 microprocessor uses instruction prefetching to improve execution speed. It fetches instructions in advance and stores them in a first-in, first-out (FIFO) instruction queue.

- The size of the instruction queue in 8086 is 6 bytes.
- This allows up to 6 bytes of instructions to be prefetched from memory before they are needed for execution.
- Prefetching reduces wait time for fetching instructions from memory, improving overall performance.

QUESTION 10:

Correct Answer: b

Detailed Solution:

The 8086 microprocessor can operate in two different modes:

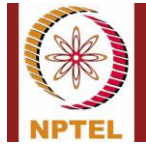
1. Minimum Mode (Single processor system)
 - Controlled by the 8086 itself.
 - No external processor or coprocessor can be attached.
2. Maximum Mode (Multiprocessor system)
 - Controlled by an external Bus Controller (8288).
 - Allows a co-processor (e.g., 8087 math co-processor) to be attached.
 - Used in high-performance applications requiring floating-point arithmetic.

QUESTION 11:

Correct Answer: b

Detailed Solution:

The given instruction:



MOV[1234h]

MOV → This is a data transfer instruction in 8086 assembly language.

[1234h] → Denotes a memory location (offset 1234h) in the current data segment (DS).

AX → The 16-bit register whose contents will be stored in memory.

Action Performed:

- The 16-bit data in AX is moved to the memory location at offset 1234h.
- The memory location [1234h] now holds the same value as AX.

QUESTION 12:

Correct Answer: a

Detailed Solution:

The WAIT instruction in the 8086 microprocessor is used to pause execution until the TEST pin (active low) is set to 0 (LOW).

- The TEST pin is typically connected to an external device.
- When executing the WAIT instruction, the CPU continuously checks the TEST pin.
- Execution resumes only when TEST = 0.

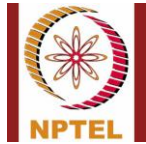
Other Options:

- BHE (Bus High Enable) → Used for accessing high-byte data in 16-bit memory operations. ✗
- DEN (Data Enable) → Used to enable external data transceivers. ✗
- HOLD → Used for DMA (Direct Memory Access) operations, not related to WAIT instruction. ✗

QUESTION 13:

Correct Answer: c

Detailed Solution:



The MOVS (Move String) instruction in 8086 is used to copy data from one memory location to another. It works with the SI (Source Index) and DI (Destination Index) registers and uses predefined segment registers for source and destination:

- Source Segment → DS (Data Segment) (SI points to the source memory location).
- Destination Segment → ES (Extra Segment) (DI points to the destination memory location).

Why ES?

- The destination address is always taken from the Extra Segment (ES).
- This allows efficient block data transfer from the Data Segment (DS) to the Extra Segment (ES).

QUESTION 14:

Correct Answer: a

Detailed Solution:

The Arithmetic Logic Unit (ALU) in the Execution Unit (EU) of the 8086 microprocessor is a 16-bit ALU.

- The 8086 is a 16-bit processor, meaning its internal registers (like AX, BX, etc.) and ALU can handle 16 bits of data at a time.
- It can perform arithmetic and logic operations (like ADD, SUB, AND, OR, etc.) on 16-bit operands.
- It also supports operations on 8-bit data, but the ALU itself is capable of 16-bit processing.
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QUESTION 15:

Correct Answer: c

Detailed Solution: In the 8086 microprocessor, the default segment register used for accessing data depends on which register is being used to form the effective address.

Default Segment Associations:

- SP (Stack Pointer) → Uses SS (Stack Segment) by default ☒



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- BP (Base Pointer) → Also uses SS (Stack Segment) by default ☒
- DI (Destination Index) → Uses DS (Data Segment) by default ☒

So:

- SP → SS ☒
- BP → SS ☒
- DI → DS ☒

*****END*****
